

1. Download these first — the core Stride dataset package

Dataset 1 — VanderPlas 37,250 marathon records

Purpose: Task 1A MLP, RQ1 and RQ2.

This is the primary benchmark containing the variables your proposal needs:

age
gender
split
final

The GitHub repository has the CSV directly and currently shows 37,251 lines including the header. ([GitHub](#))

Download page:

[VanderPlas marathon-data repository](#)

Direct CSV:

[Download marathon-data.csv directly](#)

On Ubuntu:

```
mkdir -p ~/stride/data/raw/vanderplas
```

```
cd ~/stride/data/raw/vanderplas
```

```
wget https://raw.githubusercontent.com/jakevdp/marathon-data/master/marathon-data.csv
```

You should end up with:

```
~/stride/data/raw/vanderplas/marathon-data.csv
```

2. Afonseca 10.7M running-activity dataset

Purpose: Task 1A CBR / historical training case base / RQ3.

This one is essential.

The official Figshare page currently specifies **10,703,690 records from 36,412 athletes**, with the activity data stored as Parquet files. Version 5 is the version we should use. ([Figshare](#))

Download

[Afonseca Figshare dataset — Download All](#)

The page currently shows a **510.96 MB** download and Version 5. ([Figshare](#))

Download **all files** from the Figshare page.

Store them under:

~/stride/data/raw/afonseca/

You should preserve the original Parquet files exactly as downloaded.

Important

Do **not** rename or restructure the raw files yet.

We will later produce:

athlete-week

cases from the raw activity data.

The documented variables include:

datetime
athlete
distance
duration
gender
age_group
country
major

and pre-generated daily, weekly, monthly and quarterly datasets. ([Figshare](#))

3. PhysioNet Treadmill Maximal Exercise Dataset

Purpose: Task 1B.

This is the most important additional dataset for the physiology component.

It contains:

- 992 exercise tests
- 857 participants
- 575,087 breath-level observations
- VO2
- VCO2
- HR
- RR
- VE
- treadmill speed
- age
- height
- weight
- sex
- temperature
- humidity

The current PhysioNet version is **1.0.1**. ([PhysioNet](#))

Download

[PhysioNet Treadmill Maximal Exercise Dataset v1.0.1](#)

The actual downloadable files are only about **22.1 MB uncompressed**, with a 6.3 MB ZIP. ([PhysioNet](#))

Ubuntu download

```
mkdir -p ~/stride/data/raw/physionet_treadmill
```

```
cd ~/stride/data/raw/physionet_treadmill
```

```
wget -r -N -c -np \
https://physionet.org/files/treadmill-exercise-cardioresp/1.0.1/
```

Or use the AWS command shown by PhysioNet:

```
aws s3 sync --no-sign-request \  
s3://physionet-open/treadmill-exercise-cardioresp/1.0.1/ \  
~/stride/data/raw/physionet_treadmill/
```

You need:

```
subject-info.csv  
test_measure.csv  
LICENSE.txt  
SHA256SUMS.txt
```

The dataset documentation specifically identifies `subject-info.csv` as one row per exercise test and `test_measure.csv` as the long-format breath measurements. ([PhysioNet](#))

4. Boston Marathon dataset

Purpose: External validation of Task 1A, particularly pacing/fatigue behavior.

This should **not** replace VanderPlas.

It should answer:

Does the relationship learned from the primary dataset behave similarly on another marathon/course?

The repository contains Boston Marathon data from 2001–2014, with especially detailed split information for 2013 and 2014. ([GitHub](#))

Download

[Boston Marathon 2001–2014 repository](#)

Ubuntu:

```
cd ~/stride/data/raw
```

```
git clone https://github.com/llimllib/bostonmarathon.git
```

This gives you:

```
~/stride/data/raw/bostonmarathon/
```

Do not modify the raw files.

5. AndrewMillerOnline multi-marathon dataset

Purpose: Additional external marathon-result validation.

This is worth downloading because it gives us access to substantially more race populations.

The repository currently reports **more than 2 million marathon finisher results**, including:

Berlin
Boston
Chicago
Honolulu
London
New York City
Portland

with the availability of age/splits varying by race. The repository also explicitly warns that the data have not received a complete cleaning pass. ([GitHub](#))

Download

[AndrewMillerOnline marathon-results](#)

Ubuntu:

```
cd ~/stride/data/raw
```

```
git clone https://github.com/AndrewMillerOnline/marathon-results.git
```

Important

Don't immediately throw the 2M+ records into the model.

First run:

```
race/year audit
↓
missing-value audit
↓
duplicate audit
```

↓
schema normalization
↓
select clean race/year subsets

The repository itself says the amount of information varies between races and years. ([GitHub](#))

6. Berlin Marathon dataset

Purpose: External validation and demographic/race robustness.

The repository contains CSV files for Berlin Marathon years and documents:

netTime
clockTime
yob
ageClass
sex
nationality

It currently provides API-derived CSVs from 2005 onward. ([GitHub](#))

Download

[Berlin Marathon results repository](#)

Ubuntu:

```
cd ~/stride/data/raw
```

```
git clone https://github.com/hmelberg/berlin-marathon.git
```

You will get files such as:

2005.csv
2006.csv
...
2013.csv

The repository confirms these are anonymized official Berlin results. ([GitHub](#))

7. NYC-specific 2010 dataset — optional but useful

There is one dataset I recommend downloading because of the naming issue surrounding the 37,250-row VanderPlas corpus.

The 37,250-record VanderPlas dataset should be treated as the **main marathon benchmark**, but its repository does not establish the race as NYC. ([GitHub](#))

A separate NYC Marathon dataset gives an explicitly identified **2010 New York City Marathon** cohort.

Dataset

[NYC Marathon data — CRAN/RbyExample](#)

This is small, so:

DO NOT train MLP on it.

Use it as:

NYC-specific sanity/external check

8. Wearable datasets — do NOT download all of these yet

These are useful, but they are **not required before we build the first working version of Stride**.

PAMAP2 — recommended wearable addition

This is the wearable dataset I would prioritize first.

It has:

3,850,505 instances

9 subjects

18 activities

heart rate
3 IMUs
accelerometer
gyroscope
magnetometer
temperature

and includes **running** as an activity. ([UCI Machine Learning Repository](#))

Download

[PAMAP2 — UCI Machine Learning Repository](#)

The downloadable ZIP is about **656 MB**. ([UCI Machine Learning Repository](#))

Ubuntu:

```
mkdir -p ~/stride/data/raw/pamap2
```

```
cd ~/stride/data/raw/pamap2
```

```
wget https://archive.ics.uci.edu/static/public/231/pamap2+physical+activity+monitoring.zip
```

Use this later for:

running recognition
activity segmentation
HR/intensity features
wearable preprocessing

Not marathon finish-time prediction.

9. PMData — optional

PMData is much larger than I would have you immediately download.

The official Simula page provides the full dataset as a **1.4 GB ZIP**, as well as an OSF repository. ([Simula Datasets](#))

Access

[PMData official download page](#)

[PMData OSF repository](#)

Use later for:

sleep
HR
activity
calories
recovery-related features

Priority: Medium/low.

10. Fitbit Open — optional

The Arashnic/Möbius Fitbit dataset contains data from **30 consenting Fitbit users**, including minute-level physical activity, heart rate and sleep monitoring. The current Kaggle page also shows approximately **616 MB** for the available dataset version and identifies the license as CC0. ([Kaggle](#))

Download/access

[Fitbit Fitness Tracker Data — Kaggle](#)

Use it later for:

sleep feature extraction
daily activity features
HR processing
wearable ingestion tests

Do not use it as your marathon-performance dataset.

11. PPG-DaLiA — optional, large

PPG-DaLiA is very useful for PPG/heart-rate processing, but it is **2.7 GB** and is not required for the first version of Stride. UCI lists 8.3 million instances from 15 subjects and provides the complete `data.zip`. ([UCI Machine Learning Repository](#))

Access

[PPG-DaLiA — UCI](#)

Use only when implementing:

PPG → HR

motion artifact handling

wearable signal processing

12. Sport DB 2.0 — optional

Contains:

168 cardiorespiratory datasets

130 subjects

11 sports

with ECG, HR, RR and breathing-related measurements. ([Mendeley Data](#))

Access

[Sport DB 2.0 — Mendeley Data](#)

Use for:

Task 1B physiology robustness

HR/RR validation

wearable sensor analysis

Not core MLP training.

13. PhysioNet Wearable Device Dataset — optional

This dataset provides structured wearable measurements including:

BVP

ACC

EDA
HR
IBI
temperature

The latest dataset page provides a 69.7 MB ZIP and a 247.4 MB uncompressed dataset.
([PhysioNet](#))

Access

[PhysioNet Wearable Device Dataset](#)

Terminal:

```
mkdir -p ~/stride/data/raw/physionet_wearable
```

```
wget -r -N -c -np \
https://physionet.org/files/wearable-device-dataset/1.0.1/
```

Again, this is supplementary.

14. Weather data — don't download huge archives

For Stride, I would **not download all NOAA or ERA5-Land data**.

Instead, retrieve only the weather around:

race date
race start time
race location

NOAA GHCNh

NOAA's current GHCNh is an hourly/synoptic station dataset containing temperature, humidity, wind, pressure and related observations. ([NCEI](#))

[NOAA GHCNh official data page](#)

This should become an **API/download-on-demand environmental source**, not a giant local dataset.

15. ERA5-Land — also don't download globally

ERA5-Land currently provides hourly data from 1950 onward and exposes variables including:

- 2m temperature
- 2m dewpoint
- surface pressure
- precipitation
- 10m wind U/V

The current Copernicus service provides a downloadable/queryable interface and an analysis-ready option. ([Climate Data Store](#))

[ERA5-Land official dataset](#)

For the thesis, query only:

- Boston races
- Berlin races
- NYC races

or whatever race-years you actually decide to analyze.

16. USDA FoodData Central — API, not necessarily full download

This is important for the final nutrition application, but we don't need the entire USDA database immediately.

FoodData Central provides:

- food search API
- food details API
- nutrient data

and allows downloading selected datasets as CSV/JSON. ([FoodData Central](#))

Access

[FoodData Central API Guide](#)

[FoodData Central downloadable datasets](#)

Important

You need a **data.gov API key** for normal API usage; the USDA documentation explicitly warns not to expose the key in GitHub or other public repositories. ([FoodData Central](#))

Create:

.env

with:

USDA_FDC_API_KEY=your_key_here

and add:

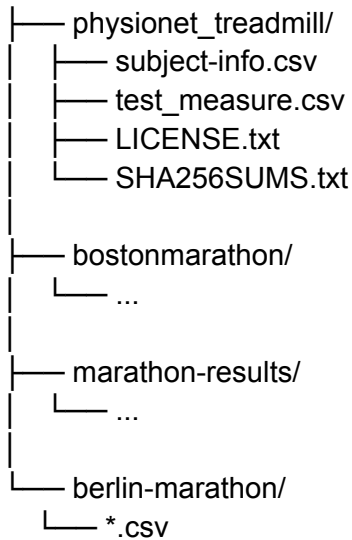
.env

to `.gitignore`.

17. The dataset package I recommend you actually download now

Rather than overwhelming yourself, your first download session should contain exactly this:

```
~/stride/
├── data/
│   └── raw/
│       ├── vanderplas/
│       │   └── marathon-data.csv
│       ├── afonseca/
│       │   └── *.parquet
│       └──
```



That is the **core research data package**.

18. What I would NOT download yet

Don't fill your computer with these before we need them:

- ✗ PPG-DaLiA — 2.7 GB
- ✗ PMData — 1.4 GB
- ✗ full NOAA historical archive
- ✗ full ERA5-Land archive
- ✗ full USDA database
- ✗ PhysioNet Wearable
- ✗ Sport DB 2.0

They are useful, but they are not blocking the first working Stride pipeline.

The first three datasets are the real blockers:

VanderPlas

Afonseca

PhysioNet Treadmill

Then:

Boston

Berlin

AndrewMillerOnline

give you stronger external validation.

19. Software/resources you should install

On your Ubuntu development machine, create a **separate Python environment for the ML work** rather than mixing it with your FastAPI environment.

Recommended:

```
cd ~/stride
```

```
python3 -m venv .venv
```

```
source .venv/bin/activate
```

Then:

```
python -m pip install --upgrade pip
```

Install the core ML/data stack:

```
pip install \
    numpy \
    pandas \
    scipy \
    scikit-learn \
    pyarrow \
    joblib \
    matplotlib \
    jupyterlab \
    openpyxl
```

I also recommend:

```
pip install duckdb polars
```

Why DuckDB is useful

Afonseca contains millions of records stored as Parquet.

Instead of loading everything into pandas at once, we can perform queries such as:

```
SELECT *
```

```
FROM read_parquet('data/raw/afonseca/*.parquet')  
LIMIT 100;
```

and build the athlete-week dataset efficiently.

For this particular project, **DuckDB + Parquet is worth having**.

20. Development tools

Have these available:

- Git
- GitHub
- VS Code
- Python
- JupyterLab
- Docker
- Postman / Insomnia
- pytest

You already have VS Code and your application backend environment, so the important addition is the isolated ML environment.

21. Python project dependencies

Eventually create:

```
requirements.txt
```

containing the actual versions used in your experiments.

At the end of the experiment, run:

```
pip freeze > requirements-lock.txt
```

Also record:

```
python --version  
pip --version
```

git --version

and:

```
python -c "import sklearn; print(sklearn.__version__)"
```

These become part of the reproducibility record.

22. Resources you need besides datasets

There are four additional things I strongly recommend preparing.

A. Research papers/reference PDFs

Keep local copies of the papers that define the important equations and nutrition assumptions:

Frayn (1983)

Minetti et al. (2002)

Uth et al. (2004)

Heikura et al. (2018)

Lukasiewicz et al. (2024)

Thomas/Erdman/Burke nutrition position statement

Feely et al. marathon recommendation work

Runtastic marathon study

These aren't training datasets, but you need them to justify the physiological implementation.

B. Dataset metadata file

Create:

data/DATASET_REGISTER.md

For every dataset:

Dataset

Source

URL

Version

Download date

License
File size
Number of records
Number of subjects/athletes
Variables
Stride role
Preprocessing
Citation
SHA-256

This becomes very useful when writing your methodology chapter.

C. Experiment registry

Create:

experiments/

with:

RQ1_mlp_vs_baseline/
RQ2_demographic_ablation/
RQ3_cbr/
RQ4_nutrition/
external_validation/
viability_test/

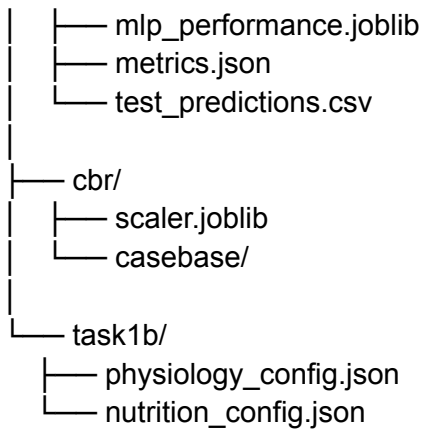
Every experiment should save:

configuration
metrics
predictions
plots
dataset version
random seed
software versions

D. Model artifact storage

Eventually:

artifacts/
|— task1a/



23. One important thing we should NOT download

Strava public dump

Remove this from your actual download list.

The current Strava API/access model does not give us a general unrestricted dataset of individual public athletes.

Therefore:

Strava



DO NOT depend on it

Our reproducible historical source is:

Afonseca

and later, for the actual application:

user-consented personal activity data

24. Final download priority

Here is the order I recommend you follow:

Priority	Resource	Download now?	Role
 1	VanderPlas 37,250	YES	Task 1A MLP, RQ1/RQ2
 2	Afonseca 10.7M	YES	Task 1A CBR, RQ3
 3	PhysioNet Treadmill	YES	Task 1B physiology
 4	Boston Marathon	YES	External pacing validation
 5	AndrewMillerOnline 2M+	YES	Multi-race external validation
 6	Berlin Marathon	YES	External validation
 7	NYC 2010 dataset	Recommended	NYC-specific sanity check
 8	PAMAP2	Later	Running/wearable features
 9	Sport DB 2.0	Later	Cardiorespiratory validation
 10	PhysioNet Wearable	Later	Wearable signal validation
 11	PMData	Later	Recovery/sleep
 12	Fitbit	Later	Wearable features
 13	PPG-DaLiA	Only if needed	PPG/HR research
 14	NOAA GHCNh	Query later	Race weather
 15	ERA5-Land	Query later	Environment
 16	USDA FoodData Central	API later	Food/nutrient composition
	Strava public dumps	NO	Not required / access unsuitable

The cleanest starting point

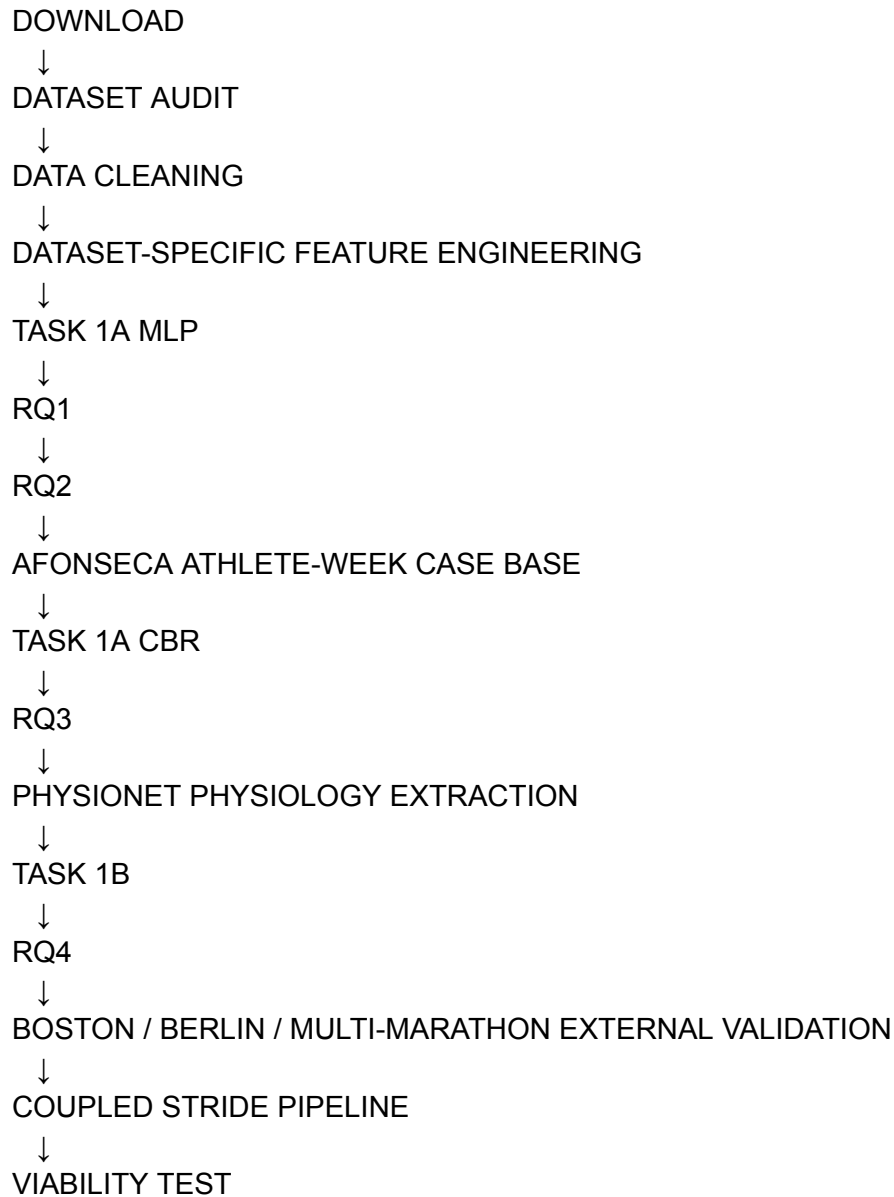
Once you have these six:

1. marathon-data.csv
2. Afonseca Parquet files
3. PhysioNet subject-info.csv
4. PhysioNet test_measure.csv
5. Boston repository

6. AndrewMillerOnline repository
7. Berlin repository

we have enough to start the **actual implementation of the ML pipeline**.

The sequence should then be:



That is the point where I would start coding rather than downloading more data.